5. Function Of Complex Variable

* complex variable :-

A complex variable is given by $z = x + iy$, where x and y are real variables and $i = \sqrt{-1}$. If z is complex variable then,

(i) The conjugate of z is given by $\bar{z} = x - iy$.

(ii) x and y are real and imaginary parts respectively of complex variable z.

* function of complex variable :-

If for each value of complex variable $z = x + iy$, in a given region R, there exist one or more values of a function $f(z) = w = u(x,y) + iv(x,y)$, where u and v are function of real variable x and y then, $f(z)$ is called as function of complex variable on a given region ~~are~~ R.

* Analytic function :-

A single valued function $f(z)$ define in a region R, is said to be analytic function if it passess a unique derivative at every point in a given region R.

The point at which the function $f(z)$ is not analytic are called singular point in that region of $f(z)$.

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* Necessary and sufficient conditions:-

The necessary and sufficient condition for $w = f(z)$ to be analytic in region R.

(i) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.

↳ Necessary cond^n

These eq^n are known as Cauchy Rieman equation.

(ii) The partial derivative $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$, $\frac{\partial v}{\partial y}$ are continuous in region R.

↳ suff^t. cond^n

* Harmonic function :-

Any function $\phi(x,y)$ of x and y having continuous partial derivative and satisfying Laplace eq^n.

then, $\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y} = 0$ is called harmonic function.

⇒ Polar form of Cauchy's Reiman eq^n :-

$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$.

Date :

Q1. ~~state~~ Test whether the following functions are analytic ?

① $f(z) = z^2$.

sol- Since, $f(z) = u + iv$

$z = x + iy$

$u + iv = (x+iy)^2$

$= x^2 + 2xyi + (iy)^2$

$= x^2 + 2xyi + i^2y^2$

$= x^2 + 2xyi - y^2$

$= x^2 - y^2 + 2xyi$

Equating,

$u = x^2 - y^2$ and $v = 2xy$

$\frac{\partial u}{\partial x} = 2x$ $\quad$ $\frac{\partial v}{\partial y} = 2x$

$\frac{\partial u}{\partial y} = -2y$ and $\frac{\partial v}{\partial x} = 2y$

$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

Hence, Cauchy's Riemann eqn are satisfied

$\therefore$ The function $f(z) = z^2$ is analytic.

② $f(z) = \bar{z}$

Sol- Since, $f(z) = u + iv$

$\bar{z} = x - iy$

Equating,

$u = x$ and $v = -y$

$\frac{\partial u}{\partial x} = 1$ $\quad$ $\frac{\partial v}{\partial y} = -1$

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$$\frac{\partial v}{\partial x} = 0$$

$$\frac{\partial u}{\partial y} = 0$$

$$\Rightarrow \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} \neq \frac{-\partial v}{\partial x}$$

Hence, Cauchy's Reiman eq^n are not satisfied.

$\therefore$ The function $f(z) = \bar{z}$ is not analytic.

(4m)

Q2. If $u = y^3 - 3x^2y$, then show that u is harmonic function.

sol- given, $u = y^3 - 3x^2y$

$$\frac{\partial u}{\partial x} = -3(2x)y = -6xy$$

$$\frac{\partial^2 u}{\partial x^2} = -6y \longrightarrow (1)$$

$$\frac{\partial u}{\partial y} = 3y^2 - 3x^2$$

$$\frac{\partial^2 u}{\partial y^2} = 6y \longrightarrow (2)$$

Adding (1) & (2),

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Hence, u is harmonic function.

Q3. If $u = e^x \cos y + x^3 - 3xy^2$, then show th u is harmonic function.

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sol- Given, $u = e^x \cos y + x^3 - 3xy^2$

$$\frac{\partial u}{\partial x} = e^x \cos y + 3x^2 - 3(1)y^2.$$

$$\frac{\partial^2 u}{\partial x^2} = e^x \cos y + 6x - 0 \longrightarrow (1)$$

$$\frac{\partial u}{\partial y} = -e^x \sin y + 0 - 6xy$$

$$\frac{\partial^2 u}{\partial y^2} = -e^x \cos y - 6x \longrightarrow (2)$$

Adding (1) & (2),

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Hence, u is harmonic function.

Q4. If $u = \frac{1}{2} \log(x^2+y^2)$, then show that u is harmonic function.

sol- Given, $u = \frac{1}{2} \log(x^2+y^2)$

$$\frac{\partial u}{\partial x} = \frac{1}{2} \cdot \frac{1}{(x^2+y^2)} \frac{\partial}{\partial x}(x^2+y^2)$$

$$= \frac{1}{2} \frac{1}{(x^2+y^2)} (2x) = \frac{x}{x^2+y^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{x}{x^2+y^2}\right] = \frac{(x^2+y^2)\frac{\partial}{\partial x}x - x\frac{\partial}{\partial x}(x^2+y^2)}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2) - x(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{y^2-x^2}{(x^2+y^2)^2} \longrightarrow (1)$$

H.W. show that $u = 2x - x^3 + 3xy^2$ is harmonic function

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$$\frac{\partial u}{\partial y} = \frac{1}{2}\frac{1}{(x^2+y^2)}\frac{\partial}{\partial y}(x^2+y^2)$$

$$= \frac{1}{2}\frac{1}{(x^2+y^2)}(2y) = \frac{y}{x^2+y^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{y}{x^2+y^2}\right) = \frac{(x^2+y^2)\frac{\partial}{\partial y}y - y\frac{\partial}{\partial y}(x^2+}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2) - y(2y)}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{x^2-y^2}{(x^2+y^2)^2} \longrightarrow (2)$$

Adding (1) & (2)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{y^2-x^2}{(x^2+y^2)^2} + \frac{x^2-y^2}{(x^2+y^2)^2}$$

$$= \frac{y^2-x^2+x^2-y^2}{(x^2+y^2)^2} = \frac{0}{(x^2+y^2)^2}$$

$$= 0$$

Hence, u is harmonic function.

Q5. Sh
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Q5. Show that $u = 2x - x^3 + 3xy^2$ is harmonic function.

Sol- Given, $u = 2x - x^3 + 3xy^2$

$$\frac{\partial u}{\partial x} = 2 - 3x^2 + 3y^2$$

$$\frac{\partial^2 u}{\partial x^2} = -6x \longrightarrow (1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} 2x - \frac{\partial}{\partial y} x^3 + \frac{\partial}{\partial y} 3xy^2$$

$$\frac{\partial u}{\partial y} = 6xy$$

$$\frac{\partial^2 u}{\partial y^2} = 6x \longrightarrow (2)$$

Adding (1) & (2)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$-6x + 6x = 0$$

Hence, u is harmonic function.

Date :

* Cauchy's Integral Theorem :-

If $f(z)$ is an analytic function and $f'(z)$ is continuous at each point within and a close simple curve C. Then,

$$\oint_C f(z)\,dz = 0$$

NOTE:- How to find poles of function

① $f(z) = \dfrac{1}{(z-2)^3}$

→ Then pole of $f(z)$ are given by,

$$(z-2)^3 = 0$$
$$(z-2)(z-2)(z-2) = 0$$
$$z = 2, 2, 2$$

$\therefore z = 2$ is pole of order 3 of $f(z)$.

Poles are inside = formula } cauchy's
Poles are outside = theorem

Date :

② $f(z) = \dfrac{3z+2}{(z-1)^2 (z+1)(z-4)}$

→ The pole of $f(z)$ are given by
$(z-1)^2 (z+1)(z-4) = 0$
$z = 1, 1$ and $z = -1; 4$

∴ The $z=1$ is pole of order 2 of $f(z)$ and $z=-1$ and $z=-4$ are simple pole of $f(z)$.

* Cauchy's Integral formula :-

If $f(z)$ is analytic function within and on a closed curve c and a is any point within c then,

$$\oint_c \frac{f(z)}{(z-a)} dz = 2\pi i f(a)$$

in general, $\oint_c \dfrac{f(z)}{(z-a)^{n+1}}$, $dz = \dfrac{2\pi i \, f^n(a)}{n!}$

∵ n = derivative

where, $f^n(z)$ denotes n^{th} derivative of $f(z)$.

NOTE :- The circle with center (x, y) and radius r.

$|z - (x+iy)| = r$

① $|z-1| = 2$
Center $(1, 0)$ and radius $= 2$

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② $|z+i| = 4$
Center $(0,-1)$ and radius $= 4$

③ $|z+3i+2| = 1$
Center $(-2,-3)$ and radius $= 1$

④ $|z| = 5$
Center $(0,0)$ and radius $= 5$

Q1. Evaluate $\oint_C \frac{z+4}{z^2+2z+5}\,dz$ where C is circle $|z+1| = 1$.

Sol- Let, $f(z) = \frac{z+4}{z^2+2z+5}$

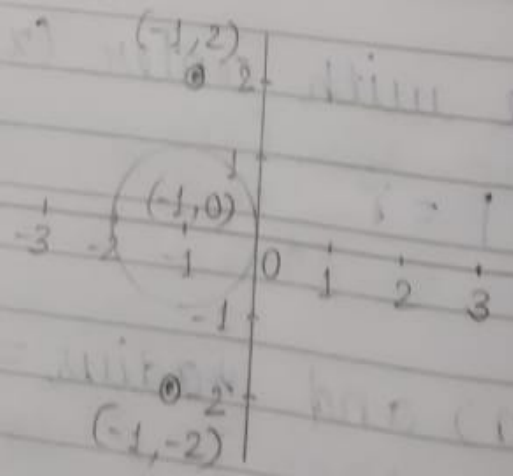
The poles of $f(z)$ are given by

$$z^2+2z+5 = 0$$

$$z = -1+2i \;,\; -1-2i$$

$z = -1+2i$ i.e, $z = (-1, 2)$
$z = -1-2i$ i.e, $z = (-1, -2)$

Since, $|z+1| = 1$ is circle with center $(-1,0)$ and radius $= 1$.



Date :

The poles $z = -1+2i$ and $z = -1-2i$ are lies outside the circle $|z+1| = 1$.

By Cauchy's integral theorem,

$$\oint_C f(z)\,dz = 0$$

$$\oint_C \frac{z+4}{z^2+2z+5}\,dz = 0$$

Q2. Evaluate $\oint_C \frac{e^{2z}}{(z+1)^4}\,dz$ where c is a circle $|z| = 3$.

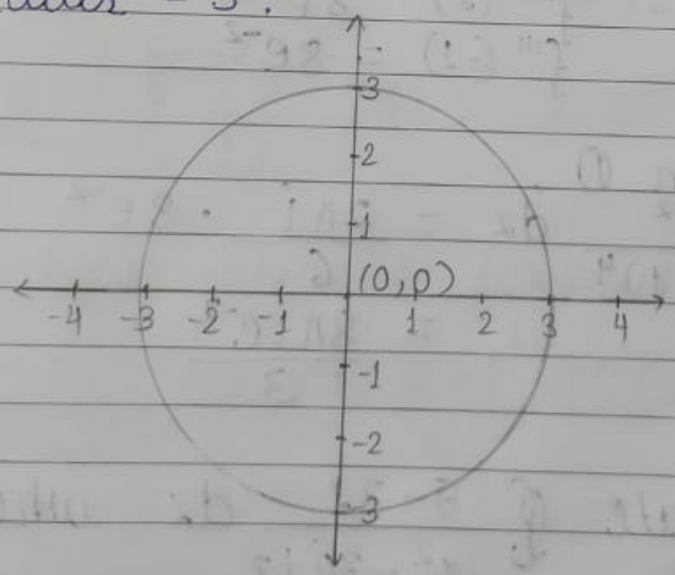
Sol- Let, $g(z) = \frac{e^{2z}}{(z+1)^4}$

The poles of $g(z)$ are given by

$$(z+1)^4 = 0$$

$$z = -1, -1, -1, -1$$

Since, $|z|=3$ is a circle with center $(0,0)$ and radius $= 3$.



The poles $z = -1$ is lies inside the circle $|z| = 3$.

By cauchy's integral formula,

$$\oint_c \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} f^n(a) \longrightarrow (*)$$

Here, $f(z) = e^{2z}$

$a = -1$

$n+1 = \cancel{3}\,4 \Rightarrow n = 3$

from (*),

$$\oint_c \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i}{3!} f'''(-1) \longrightarrow (1)$$

Since, $f(z) = e^{2z}$

$f'(z) = 2e^{2z}$

$f''(z) = 4e^{2z}$

$f'''(z) = 8e^{2z}$

$f'''(-1) = 8e^{-2}$

from (1),

$$\oint \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i}{6} \cdot 8e^{-2}$$

$$= \frac{8\pi i e^{-2}}{3}$$

3. Evaluate $\oint_c \frac{5-3z}{z^2-3z+2} dz$ where c is circle $|z| = \frac{3}{2}$.

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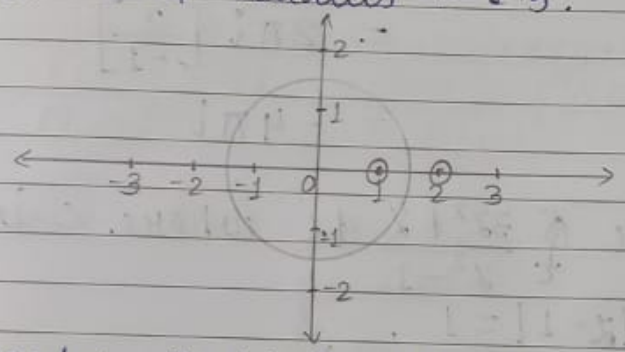
Sol- Let, $g(z) = \dfrac{5-3z}{z^2-3z+2}$

The poles of $g(z)$ are given by.

$$z^2-3z+2 = 0$$
$$z = 2, 1$$

Since, $|z| = 3/2$ is circle with center $(0,0)$ and radius $= 1.5$.



The poles $z=1$ as lies inside and $z=2$ lies outside of circle $|z| = 3/2$.

given, $\oint_c \dfrac{5-3z}{z^2-3z+2}\, dz = \oint_c \dfrac{5-3z}{(z-1)(z-2)}\, dz$

$$= \oint_c \frac{\left(\frac{5-3z}{z-2}\right)}{(z-1)}\, dz$$

By cauchy's integral formula,

$$\oint_c \frac{f(z)}{(z-a)^{n+1}}\, dz = \frac{2\pi i}{n!} f^n(a) \quad \rightarrow (*)$$

Here, $f(z) = \dfrac{5-3z}{z-2}$

$\boxed{0! = 1}$

$a = 1$
$n+1 = 1 \Rightarrow n = 0$

from ⊛,

$$\oint_c \frac{\left(\frac{5-3z}{z-2}\right) dz}{(z-1)} = \frac{2\pi i}{0!} f(1)$$

$$= \frac{2\pi i}{1} \left[\frac{5-3(1)}{1-2}\right]$$

$$= 2\pi i \left[\frac{2}{-1}\right]$$

$$= -4\pi i$$

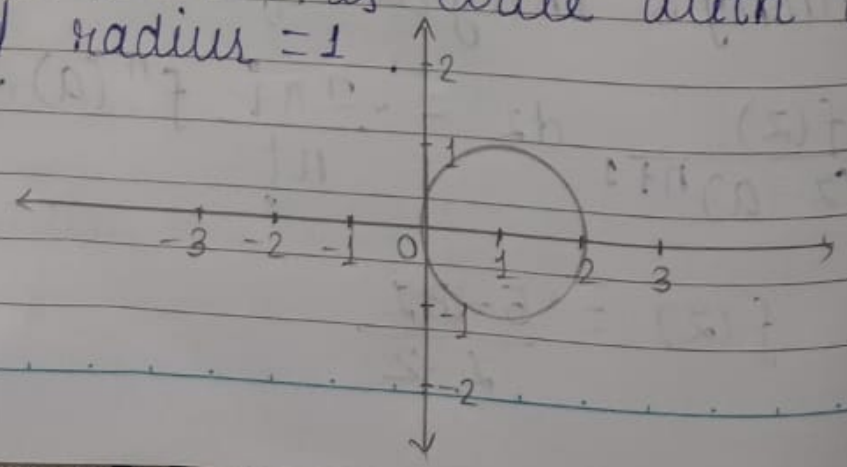
Q4. Evaluate $\oint_c \frac{3z^2+z}{z^2-1} dz$ where c is a circle $|z-1| = 1$.

Sol- Let, $g(z) = \oint_c \frac{3z^2+z}{z^2-1} dz$

The poles of $g(z)$ are given by

$$z^2 - 1 = 0$$
$$z^2 = 1$$
$$z = \pm 1$$
$$z = 1, -1.$$

since $|z-1| = 1$ is circle with cente and radius $= 1$.



$\cos \pi = -1$

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The pole $z=1$ lies inside and $z=-1$ lies outside the circle $|z-1| = 1$

given, $\oint_C \frac{3z^2+z}{z^2-1}\, dz = \oint_C \frac{3z^2+z}{(z-1)(z+1)}\, dz$

$$= \oint_C \frac{\left(\frac{3z^2+z}{z+1}\right)}{(z-1)}\, dz$$

By using Cauchy's integral formula

$$\oint_C \frac{f(z)}{(z-a)^{n+1}}\, dz = \frac{2\pi i}{n!} f^{n}(a) \longrightarrow (*)$$

Compare, $f(z) = \frac{3z^2+z}{z+1}$

$a = 1$

$n+1 = 1 \Rightarrow n = 0$

from $(*)$

$$\oint_C \frac{\left(\frac{3z^2+z}{z+1}\right)}{z-1}\, dz = \frac{2\pi i}{0!} f(1)$$

$$= \frac{2\pi i}{1}\left[\frac{3(1)^2+1}{1+1}\right]$$

$$= 2\pi i \left(\frac{4}{2}\right) = 4\pi i$$

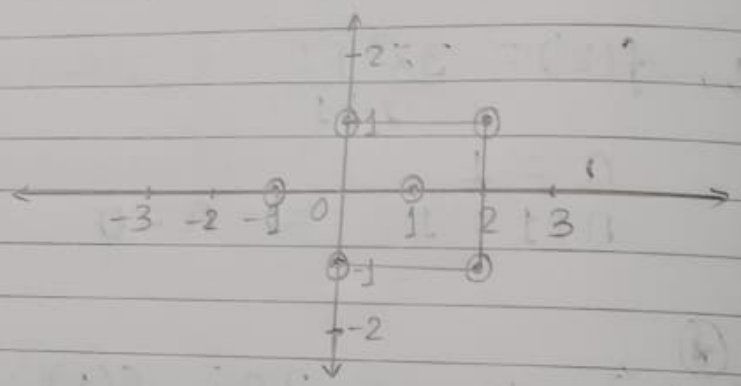
Evaluate $\oint_C \frac{\cos \pi z}{z^2-1}\, dz$ around a rectangle with vertices $2 \pm i$, $\pm i$.

Sol- Let, $g(z) = \frac{\cos \pi z}{z^2 - 1}$

The poles of $g(z)$ are given by.

$$z^2 - 1 = 0$$
$$z^2 = 1$$
$$z = \pm 1$$
$$z = 1, -1$$

Since, rectangle with vertices
$2 \pm i$, i.e, $(2,1)$, $(2,-1)$
$\pm i$, i.e, $(0,1)$, $(0,-1)$



The pole $z=1$ lies inside and $z=-1$ lies outside the rectangle.

Given, $\oint_C \frac{\cos \pi z}{z^2-1} dz = \oint_C \frac{\cos \pi z}{(z+1)(z-1)} dz$

$$\oint_C \frac{\cos \pi z}{z^2-1} dz = \oint_C \frac{\left(\frac{\cos \pi z}{z+1}\right)}{z-1} dz$$

By using cauchy's formula,

Date :

$$\oint_c \frac{f(z)}{(z-a)^{n+1}}\,dz = \frac{2\pi i}{n!} f^n(a) \longrightarrow (*)$$

Compare, $f(z) = \dfrac{\cos \pi z}{z+1}$

$a = 1$

$n+1 = 1 \Rightarrow n = 0$

from $(*)$

$$\oint \frac{\left(\frac{\cos \pi z}{z+1}\right)}{z-1}\,dz = \frac{2\pi i}{0!} f(1)$$

$$= \frac{2\pi i}{1} \frac{\cos \pi(1)}{1+1}$$

$$= 2\pi i \frac{(-1)}{2} = -\pi i$$